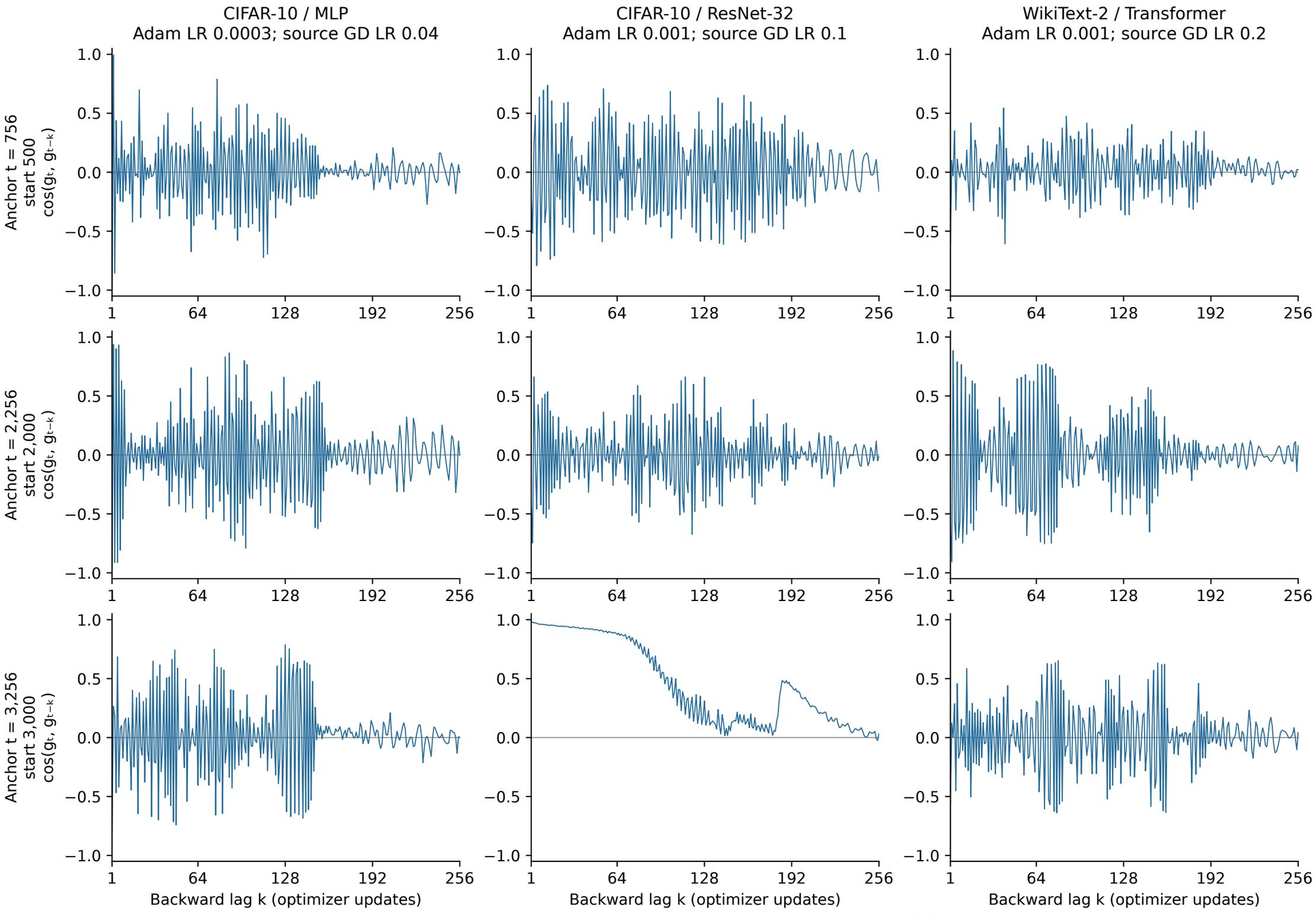

Backward anchored raw-gradient cosine along Adam
CIFAR-10 / MLP
Adam LR 0.0003; source GD LR 0.04
CIFAR-10 / ResNet-32
Adam LR 0.001; source GD LR 0.1
WikiText-2 / Transformer
Adam LR 0.001; source GD LR 0.2
Anchor t = 756
start 500
cos(gt, gt−k)
Anchor t = 2,256
start 2,000
cos(gt, gt−k)
Anchor t = 3,256
start 3,000
cos(gt, gt−k)
1.0
0.5
0.0
−0.5
−1.0
1
64
128
192
256
Backward lag k (optimizer updates)
Fixed endpoint t = retained start + 256; compare gt with gt−k for every k = 1–256.